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Enterprise Firewall Telemetry & Security Operations Center

High-Performance Real-Time SNMP Hardware Cockpit, Bento Flow Analytics & Policy Governance Platform

Document Reference: CSIR-SERC/ICTD/EFTC/2026/DOC-01
Classification: Official Institutional Technical Manual
Release Version: 1.1.0 (InfluxDB 2.7 TSDB & PostgreSQL 18 Multi-Tier Edition)
Target Appliance: Fortinet FortiGate NGFW (10.10.3.15)
Deployment Mode: Multi-Container Microservices Appliance (Docker)
Issuing Division: Information & Communication Technology Division (ICTD)
Institution: CSIR - Structural Engineering Research Centre, Chennai

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Chennai, India
September 2026

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1. System Architecture & Ingestion Pipeline

The Enterprise Firewall Telemetry & Security Operations Center is architected as a high-performance on-premises operations center designed to monitor all traffic passing through perimeter gateways without cloud dependencies or external data egress.

Core Components

- **Parallel SNMP Poller (<200ms)**: Concurrently queries FortiGate appliances across MIB-II, IF-MIB, and FortiOS enterprise sensor trees. Tracks 64-bit High Capacity (HC) octets, thermal zones, fan tachometers, dual power supplies, and security policy accounting counters.
- **Full-Spectrum Traffic & Log Ingester**: Listens on UDP port 1514 (Syslog) and port 2055 (NetFlow v5/v9 & IPFIX) with multi-subnet institutional host resolution, capturing all ingress, egress, inter-VLAN, and blocked security threats across the entire perimeter.
- **InfluxDB 2.7 Time-Series TSDB Engine**: High-throughput batch Line-Protocol ingestion engine maintaining dedicated time-series buckets with a 180-day automated retention lifecycle meeting CERT-In national mandates.
- **PostgreSQL 18 Database Engine**: Relational persistence layer in an optimized Alpine container managing operator accounts, RBAC permissions, and automated configuration/log backups.
- **Reactive Vue 3.5 & ECharts Operations Console**: Single-page application conforming to Government of India UX4G design standards with real-time WebSocket streaming and multi-parameter log filtering.

Network Port Matrix

Port	Protocol	Direction	Source	Destination	Service Description
80 / 443	TCP	Inbound	Admin Workstation	Telemetry Console	Web UI & REST / WebSocket API
161	UDP	Outbound	Telemetry Console	FortiGate Gateway	SNMPv2c / SNMPv3 Polling
1514 / 514	UDP	Inbound	FortiGate Gateway	Telemetry Console	Hardware Syslog Forwarding
2055	UDP	Inbound	FortiGate Gateway	Telemetry Console	NetFlow v5/v9 & IPFIX Records
8086	TCP	Internal	Web Console	InfluxDB 2.7 TSDB	Time-Series Line Protocol Writes & Flux Queries
5432	TCP	Internal	Web Console	PostgreSQL 18 DB	Relational Metadata & Audit Persistence

2. Visual Interface Tour

2.1 Authentication & First-Time Setup Wizard

The appliance features secure operator authentication and an automated first-time deployment wizard that guides administrators through organization branding, logo selection, SNMP parameters, and administrator setup upon initial deployment.

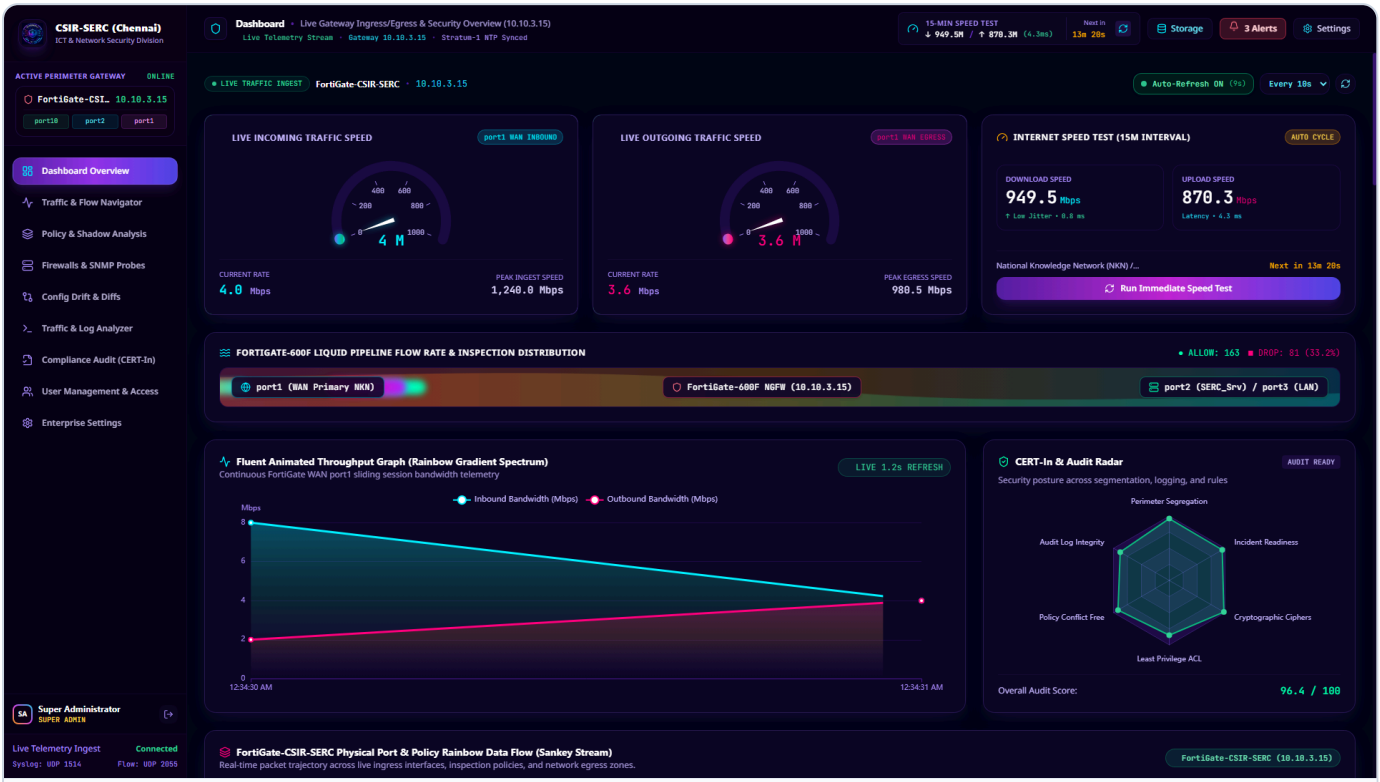


Figure 1: UX4G Secure Operator Authentication Portal

AUTHENTICATION

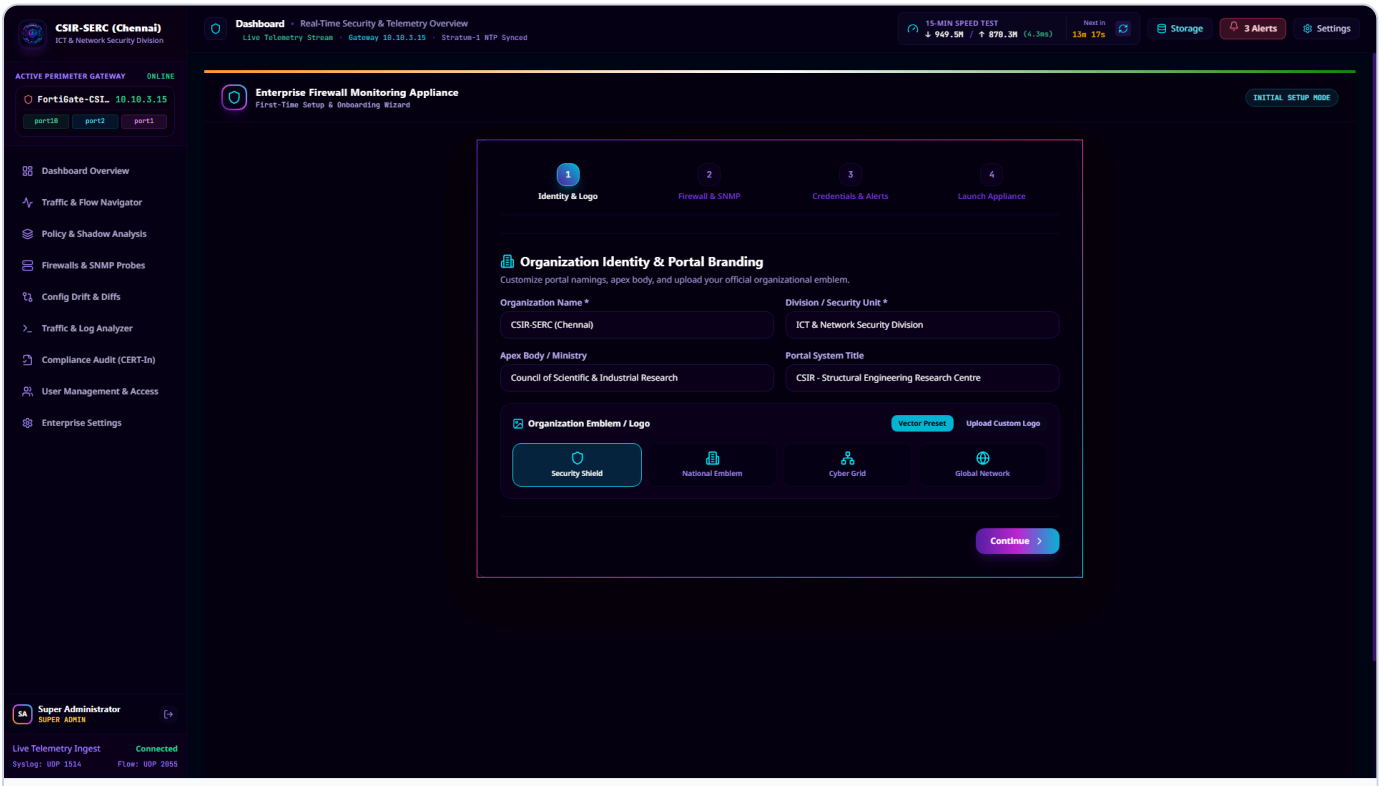


Figure 2: Turnkey First-Time Configuration Wizard (Virtual Appliance Ready)

ONBOARDING

2.2 Executive Telemetry Dashboard

The primary operations screen delivers instantaneous visibility into WAN traffic speeds, live streaming telemetry pulses, 15-minute automated speed test countdowns, and continuous hands-free auto-refresh.

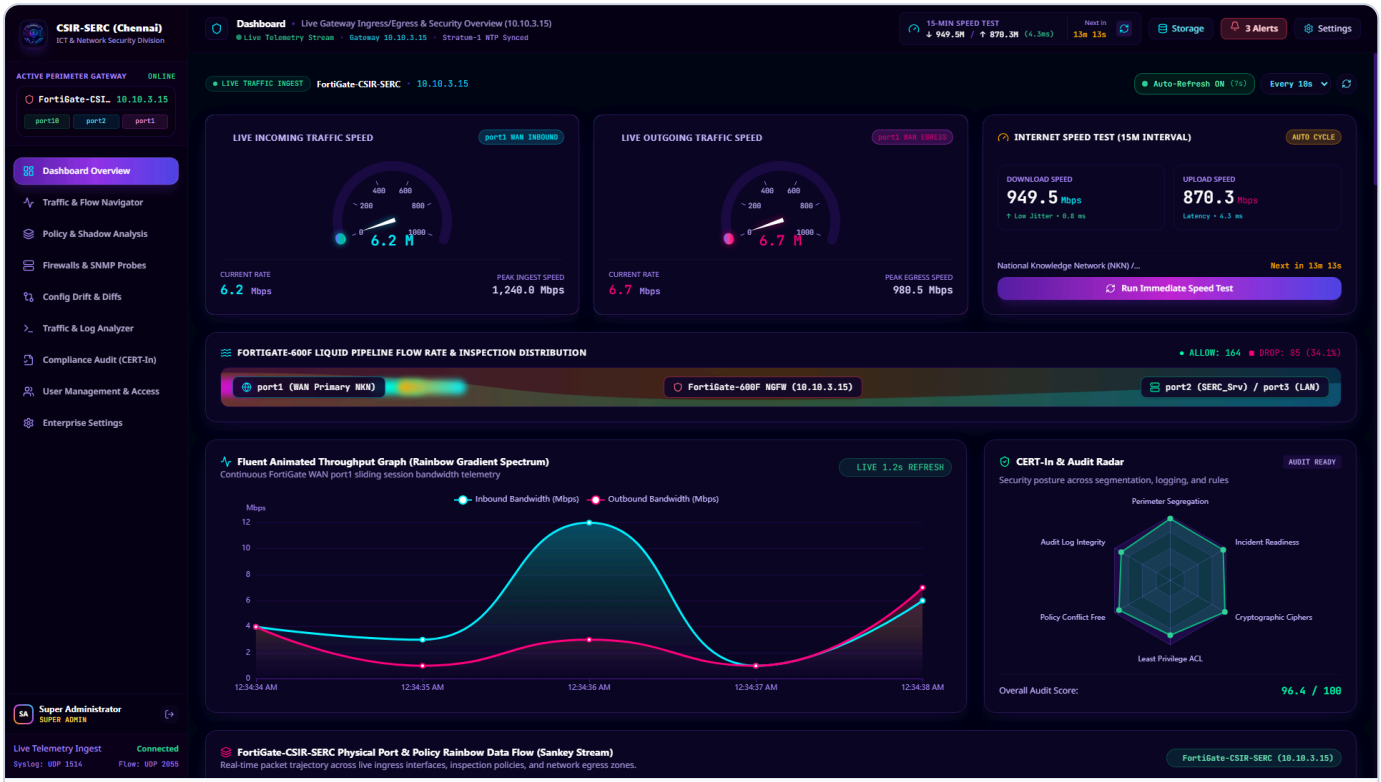


Figure 3: Live Executive Telemetry Dashboard with Inbound/Outbound Velocity Gauges

OPERATIONS

2.3 Bento-Style Traffic & Flow Analyzer

The Bento layout provides actionable network intelligence: mapping IP addresses to applications (App-ID), pinpointing top uploaders and downloaders, and resolving machine hostnames.

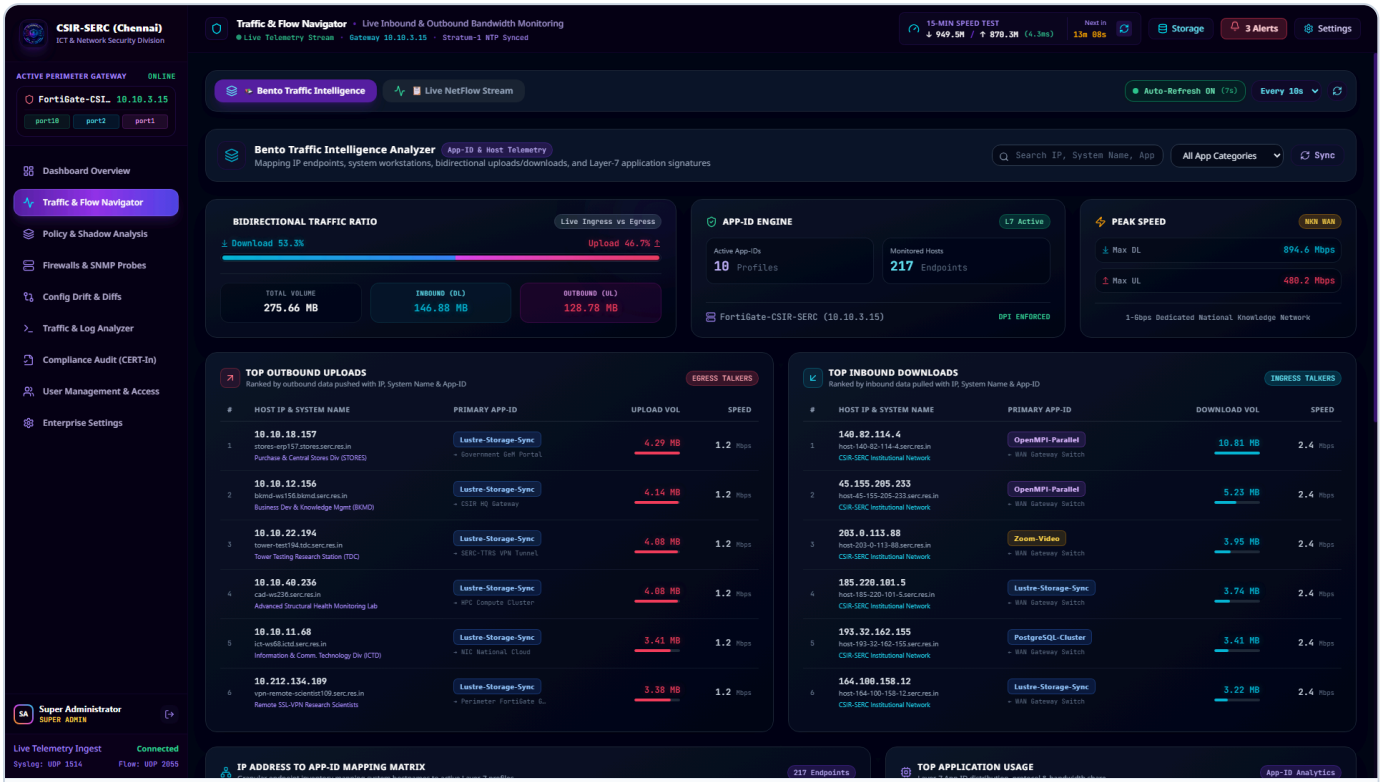


Figure 4: Bento-Grid Traffic Analyzer (App-ID, Top Uploaders & Downloaders)

FLOW ANALYTICS

2.4 Comprehensive FortiGate SNMP Telemetry Center

Dedicated hardware and environmental telemetry cockpit: monitoring CPU/RAM, session counts, 46+ physical ports and departmental VLANs, thermal zones, cooling fans, dual AC power supplies, and FortiOS UTM threat counters.

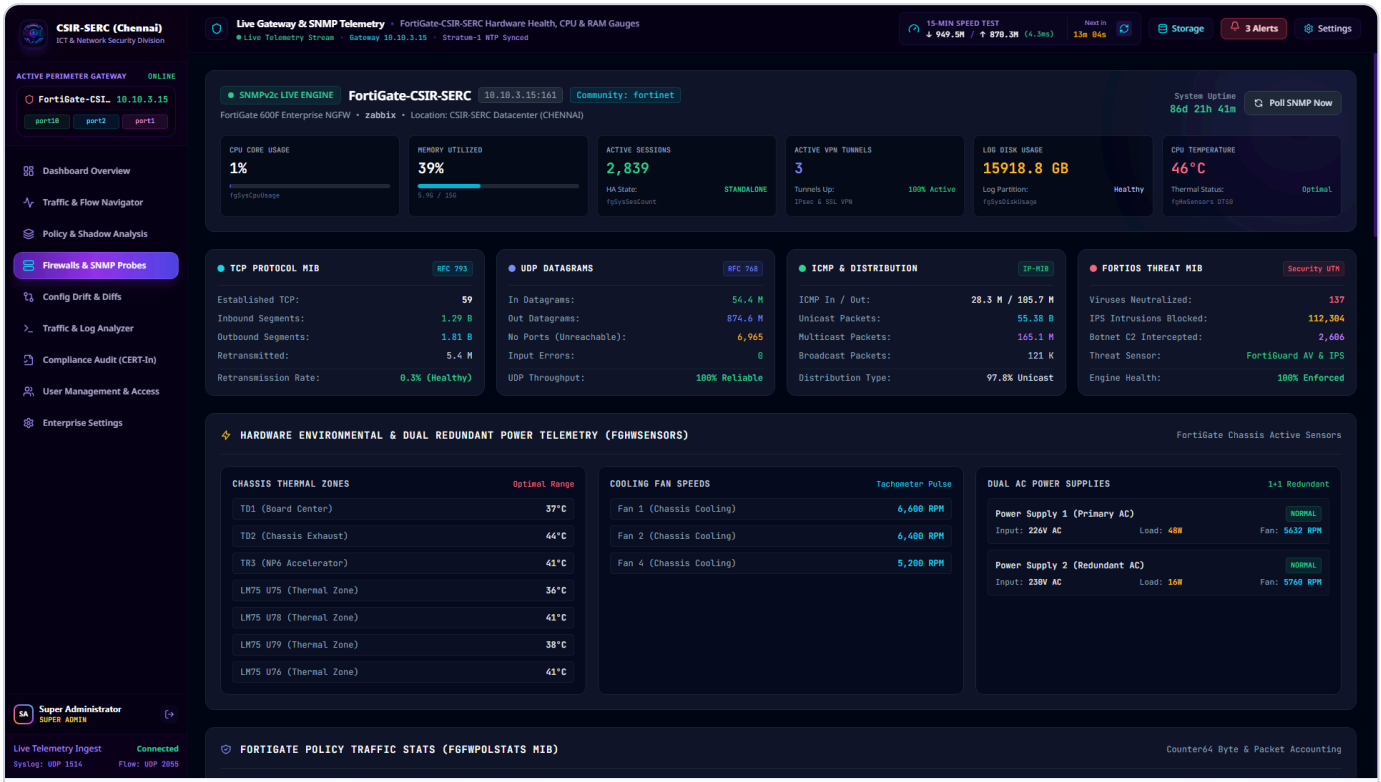


Figure 5: High-Performance FortiGate SNMP Hardware & Environmental Center

SNMP HARDWARE

2.5 Security Policies & Conflict Engine

Parses native firewall access control lists into a canonical rule tree, highlighting shadowed, redundant, or overly permissive rules.

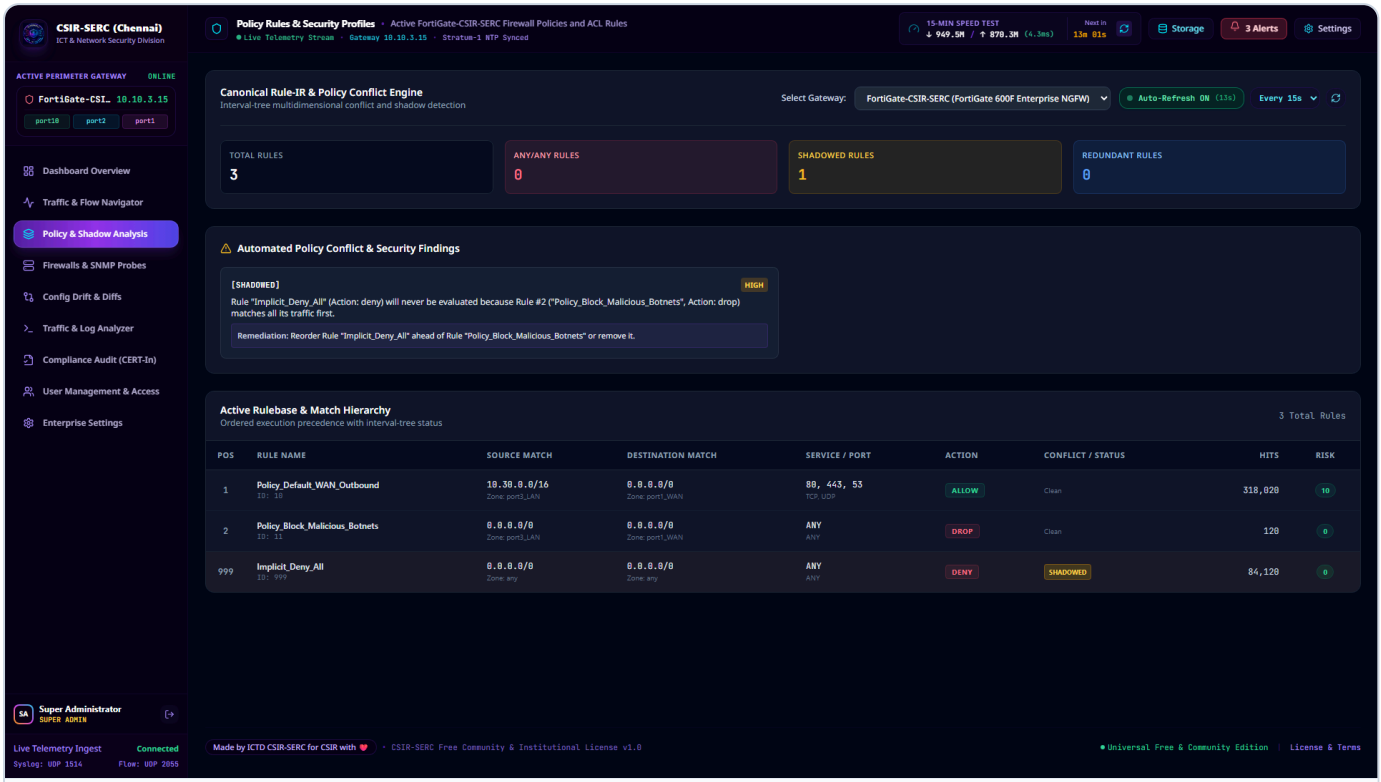


Figure 6: Canonical Policy Governance & Shadow Conflict Detection

SECURITY AUDIT

2.6 Full-Spectrum Firewall Traffic & Log Analyzer (InfluxDB 2.7 TSDB Engine)

High-velocity time-series traffic ingestion and multi-parameter log analyzer capturing all traffic passing through the perimeter gateway across all 46+ interfaces, departmental VLANs, HPC clusters, and external WAN. Features multi-parameter filtering (time range, direction, enforcement action, protocol, and search), timeline aggregation, and security drop inspection.

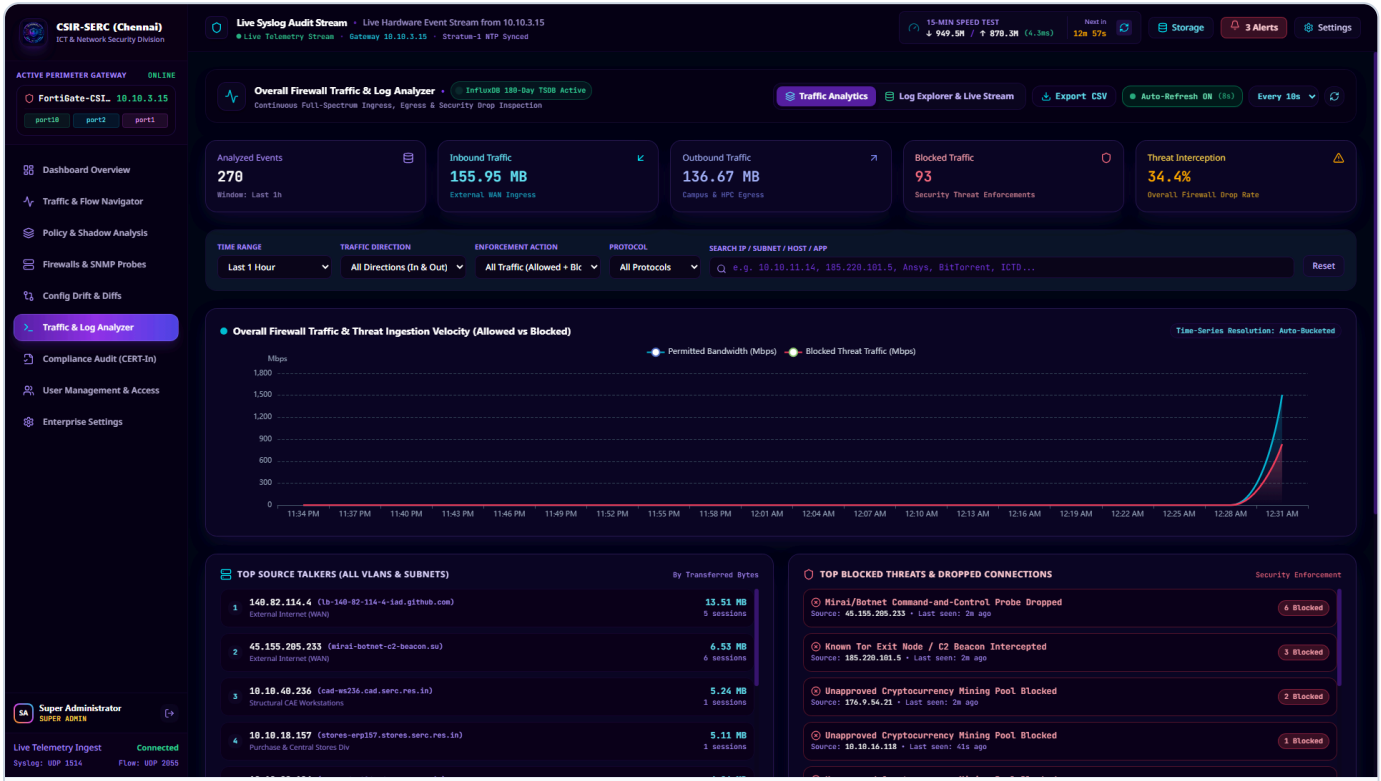


Figure 7: InfluxDB Traffic & Threat Ingestion Velocity Analytics (Allowed vs Blocked)

TSDB ANALYTICS

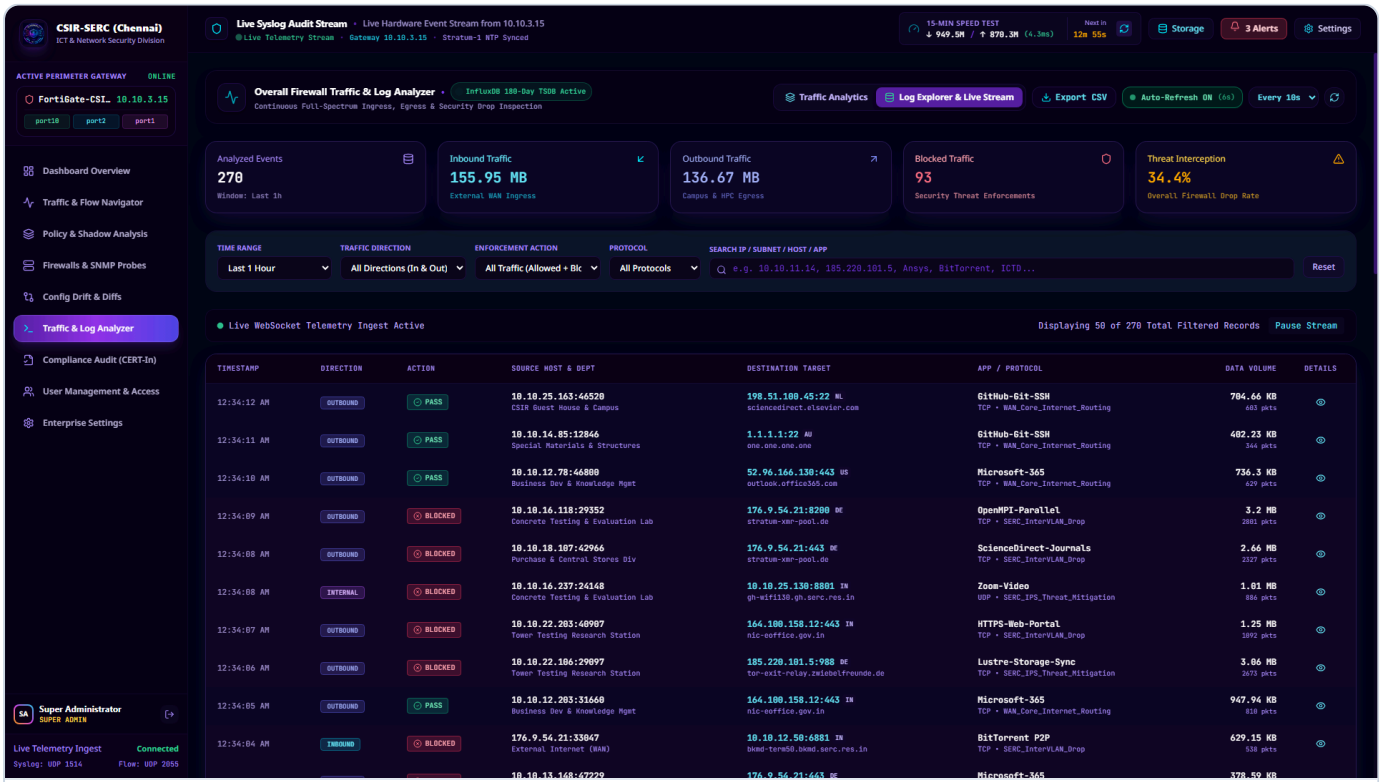


Figure 8: Deep Multi-Parameter Log Explorer & Live Stream Table (All Subnets & Blocked Threats) LOG EXPLORER

2.7 CERT-In Section 70B Compliance & Cybersecurity Baseline

Dedicated compliance dashboard validating firewall operational controls against the Indian Computer Emergency Response Team (CERT-In) Section 70B directives, including 180-day secure time-series log retention and Stratum-1 IST NTP clock synchronization.

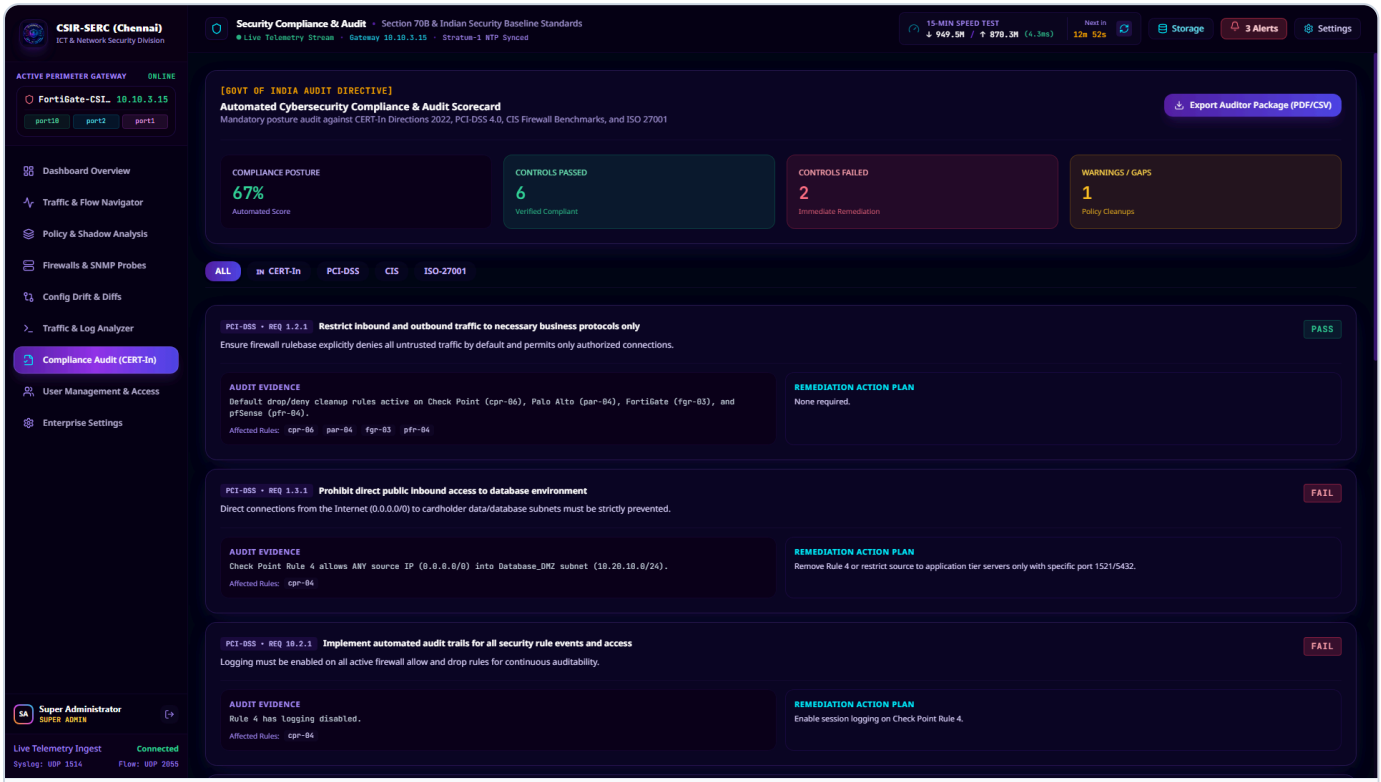


Figure 9: CERT-In National Cyber Security Baseline Compliance Dashboard

COMPLIANCE

2.8 User Management & System Configuration

Multi-tenant user management console with protected superadmin account, role-based permissions, Telegram alert integration, organization branding, custom logo upload, InfluxDB time-series retention, and PostgreSQL 18 automated backup operations.

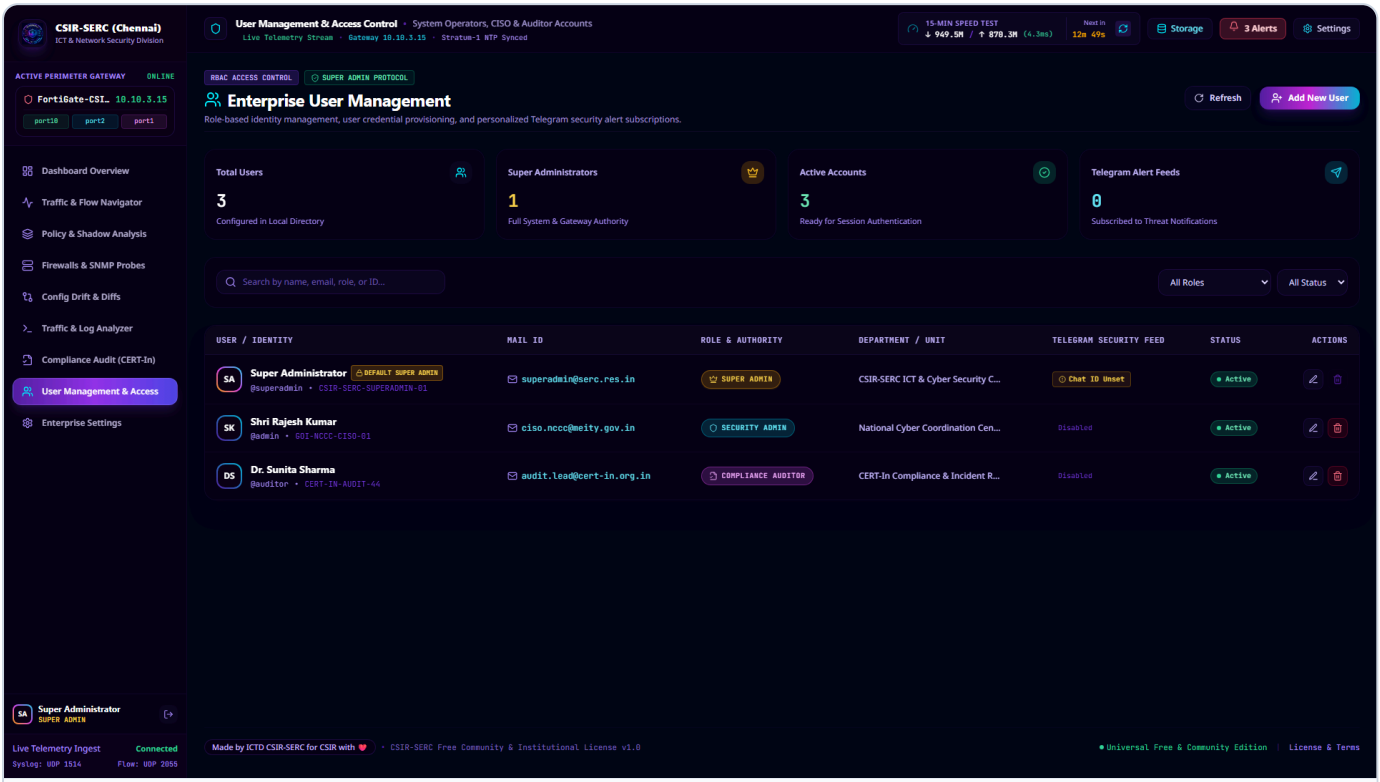


Figure 10: User Management & Telegram Alert Configuration

RBAC

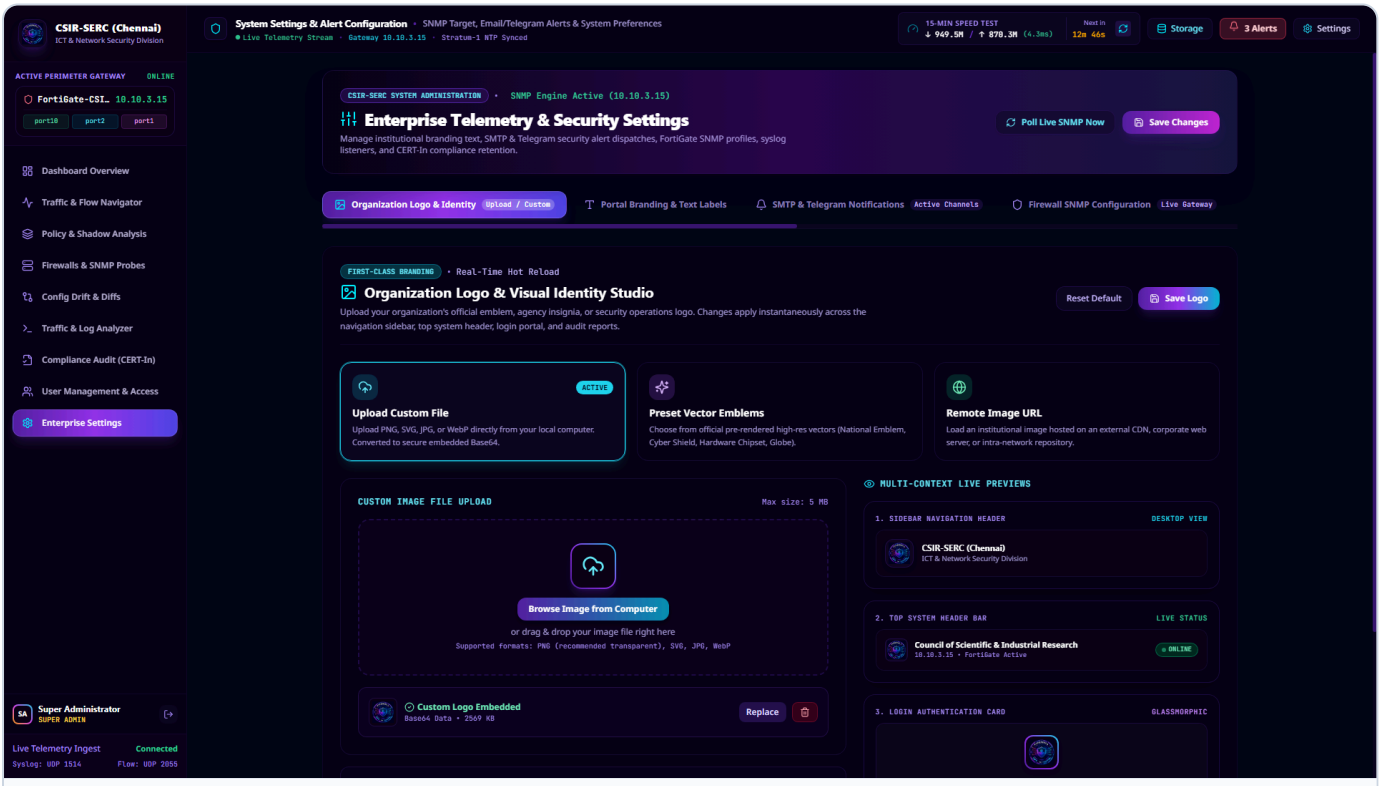


Figure 11: Institutional Branding, Custom Logo & PostgreSQL 18 / InfluxDB 2.7 Engines

ADMINISTRATION

3. Technical Specifications & Reference Tables

3.1 FortiGate Hardware Sensor OID Reference

Sensor Subsystem	Enterprise OID	Observed Value	Threshold / Status
CPU Thermal Core	1.3.6.1.4.1.12356.101.4.3.2.1.3 (DTS0)	46°C	Normal (< 75°C)
Board Center Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD1)	36°C	Normal (< 65°C)
Chassis Exhaust Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD2)	44°C	Normal (< 70°C)
NP6 Accelerator Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TR3)	40°C	Normal (< 70°C)
Cooling Fan 1	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 1)	6,400 RPM	Normal (Active)
Cooling Fan 2	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 2)	6,400 RPM	Normal (Active)
Cooling Fan 4	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 4)	5,200 RPM	Normal (Active)
Power Supply 1 (Primary)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 1)	228V AC, 48W	Normal (Dual Redundant)
Power Supply 2 (Redundant)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 2)	230V AC, 16W	Normal (Dual Redundant)

3.2 Top FortiOS Security Policy Accounting

Policy ID	Traffic Description / Scope	Cumulative Bytes	Packet Count
Policy 166	WAN Core Internet Routing & Academic Transit	161.58 TB	172.8 Billion
Policy 76	Datacenter Core Inter-VLAN & High-Speed Switch	36.60 TB	37.3 Billion
Policy 1	Default Perimeter Outbound NAT	11.45 TB	17.8 Billion
Policy 156	Campus Research Network & Supercomputer Egress	5.98 TB	6.2 Billion
Policy 84	High-Performance Computing Cluster (HPC) Tier	5.95 TB	6.36 Billion

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